

Intertemporal Choice

JEB104 Microeconomics I
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Present and Future Values

Assumptions:

- Two periods of time: 1 and 2
- Money can be saved in the period 1 at the interest rate r .
- Example: if $r = 0.1$, \$100 dollars saved in period 1 yield \$110 in period 2.

Future value

- the next-period value of the m saved now

$$FV = m(1 + r)$$

Present value

- the present value of m received in the next period

$$PV = \frac{m}{1 + r}$$

Previously

- Analysis was made in one time period, we did not analyze any past or future periods.
- We assumed that the consumer spent his entire income in this given period.

Now: Analyze how consumer makes his decisions about consumption over time.

- There are more periods.
- Consumer can save part of his income for future, or borrow some money which should be repaid in future.

Future value

- Trivial, everyone knows the situation - you put some money into a bank, next year you have money plus interest.

Present value

- How we value m received in period 2 now in period 1?
- Suppose you can pay now to obtain \$1 at the start of the next period.
- How much should you pay? What is the most you should pay? \$1? No.
- If you kept your \$1 now and saved it then at the start of the next period you would have $\$(1 + r) > \1 , so paying \$1 now for \$1 next period is a bad deal.
- So the present value expresses how much money would have to be saved now, in the present, to obtain \$1 at the start of the next period.
- $x = 1/(1 + r)$: the present-value of \$1 (\$1 is obtained at the start of the next period).

Intertemporal Choice

Assumptions:

- m_1 and m_2 are incomes received in periods 1 and 2;
- c_1 and c_2 are the amounts of consumption in periods 1 and 2;
- prices of consumption goods in each period $p_1 = p_2 = 1$.

The intertemporal choice problem:

Given incomes m_1 and m_2 , what is the most preferred intertemporal consumption bundle (c_1, c_2) ?

We need to know:

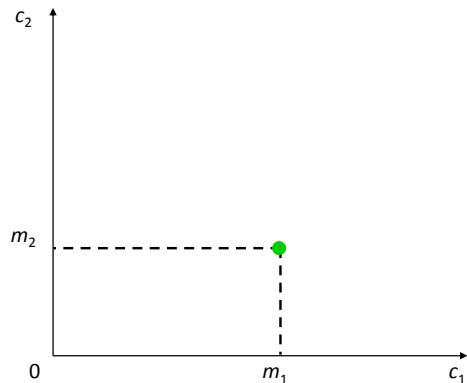
- budget constraint
- intertemporal preferences

Prices are 1: composite good – dollars spent on consumption. We assume that prices (price level) do not change over time.

How incomes in period 1 and period 2 will be transformed to consumption in different periods?

Budget Constraint

Consumer cannot save or borrow.



$$c_1 = m_1, c_2 = m_2$$

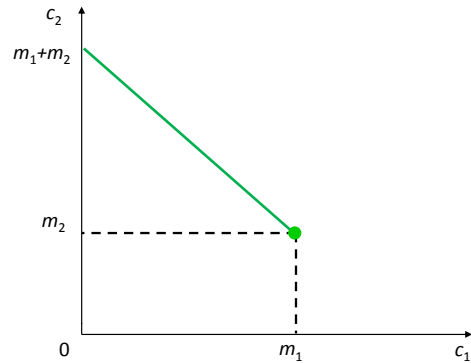
Let me know describe several budget constraints which can be possible in this setting.

First case is straightforward.

- If you cannot save or borrow, you need to spend your full income in each period.
- In this case, we are back in our baseline analysis. Income is fully spent in a given period.

Budget Constraint

Consumer can save income from the period 1 without earning interest.



Zero interest rate, no borrowing allowed.

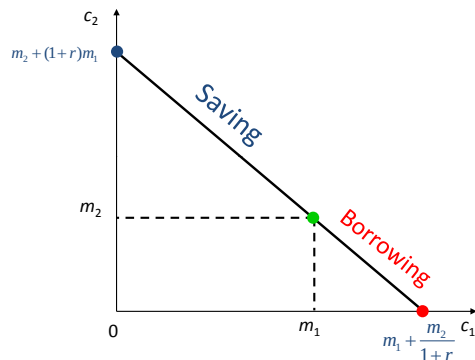
$$c_1 + c_2 \leq m_1 + m_2, \quad c_1 \leq m_1$$

Budget Constraint

Consumer can save or borrow at interest rate r .

If $c_1 = 0 \Rightarrow c_2 = (1 + r)m_1 + m_2$ (Consumer saves all m_1 for period 2)

If $c_2 = 0 \Rightarrow c_1 = m_1 + \frac{m_2}{1 + r}$ (Consumer borrows all m_2 and repays in period 2)



Future value of income endowment: $m_1(1 + r) + m_2$

Present value of income endowment: $m_1 + \frac{m_2}{1 + r}$

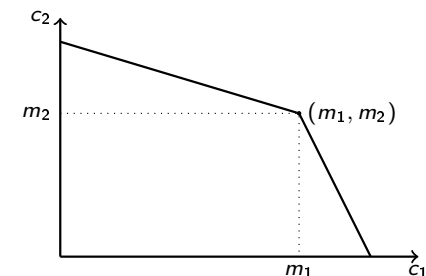
Consumer can save, but saved money does not earn any interest.

- Money is simply transferred from period 1 to period 2 (if you want).
- You just put part of your income from period 1 aside.
- Slope of the budget constraint is -1 , \$1 saved in period 1 gives you \$1 dollar in period 2.

This is the budget constraint we will consider in our further analysis.

Saving and borrowing:

- Consumer can save income from the period 1 at interest rate r .
- Consumer can borrow money in the period 1 at interest rate r which will be repaid in the period 2.
- We assume that interest rate r is the same for savings and borrowings. In reality, it is not the case. Generally, your savings earn lower interest than what you have to pay for your borrowings.
- Keeping the interest rate the same for savings and borrowings makes the analysis simpler. But we can adapt the analysis also for different interest rates for savings and borrowings. In such case, convexity of the budget set would be preserved. The part of the budget line where savings are realized would be flatter than the other part of the budget line (borrowings).



Budget Constraint

If m_1 is income and c_1 consumption in period 1, then:

$$c_2 = m_2 + (m_1 - c_1) + r(m_1 - c_1)$$

$$c_2 = \underbrace{-(1+r)}_{\text{slope of BC}} c_1 + \underbrace{m_2 + m_1(1+r)}_{\text{vertical intercept}}$$

Future-valued form of BC:

$$c_1(1+r) + c_2 \leq m_1(1+r) + m_2$$

Present-valued form of BC:

$$c_1 + \frac{c_2}{1+r} \leq m_1 + \frac{m_2}{1+r}$$

Budget Constraint

Let's consider different prices for consumption in periods 1 and 2, p_1 and p_2 .

Maximum possible spending in period 2: $m_2 + (1+r)m_1$

Maximum possible consumption in period 2: $c_2 = \frac{m_2 + (1+r)m_1}{p_2}$

Maximum possible spending in period 1: $m_1 + \frac{m_2}{1+r}$

Maximum possible consumption in period 1: $c_1 = \frac{m_1 + m_2/(1+r)}{p_1}$

If c_1 consumed in period 1 $\Rightarrow p_1 c_1$ spent in period 1, leaving $m_1 - p_1 c_1$ saved for period 1. Available income in period 2:

$$m_2 + (1+r)(m_1 - p_1 c_1)$$

Consumption in period 2:

$$p_2 c_2 \leq m_2 + (1+r)(m_1 - p_1 c_1)$$

The equation for c_2 is intuitive for $m_1 > c_1$ (savings in period 1). If $c_1 > m_1$ (consumer is a borrower in period 1), then:

$$c_2 = m_2 - (c_1 - m_1) - r(c_1 - m_1)$$

Both expressions are, however, equivalent and lead to the same equation of the budget line.

We can write the budget constraint in terms of future value or in terms of present value. It does not matter which one you choose. Let us compare these equations with the traditional form $p_1 x_1 + p_2 x_2 \leq m$.

Future-valued form of BC

- Price in the future period as 1, while price in the current period is $1+r$ (period-1 price measured relative to the period-2 price)
- Unit of measurement of value income and consumption is 1 dollar in future (hence price of current consumption in dollars in future is $1+r$)

Present-valued form of BC

- More important / it measures future relative to the present (this is way we naturally look at it, because we plan our consumption now in the current period)
- unit of measurement is now 1 dollar in current period (hence price of future consumption in current dollars is $1/(1+r)$, 1 dollar in future has value $1/(1+r)$ now)

Now let us also discuss the case when we have different price levels in different periods.

- Prices usually increase, hence $p_2 > p_1$ due to inflation. So for \$1 you can buy less consumption in period 2 than in period 1.

We will study how introducing different price levels changes the budget constraint.

The expression of income available in period 2 is again intuitive for savings. If $p_1 c_1 > m_1$ and consumer is a borrower in the period 1, the income available in period 2 would be:

$$m_2 - [(p_1 c_1 - m_1)(1+r)]$$

Both expressions are, however, equivalent.

Budget Constraint

Future-valued form of BC:

$$(1+r)p_1c_1 + p_2c_2 \leq m_1(1+r) + m_2$$

Present-valued form of BC:

$$p_1c_1 + \frac{p_2c_2}{1+r} \leq m_1 + \frac{m_2}{1+r}$$

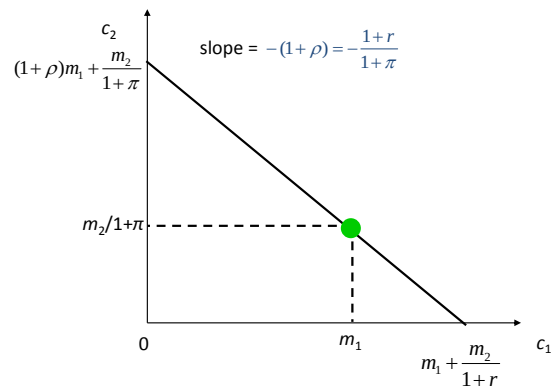
If $p_1 = 1$ and inflation rate is π , then $p_2 = 1 + \pi$. Budget constraint is then:

$$c_1 + \frac{1+\pi}{1+r}c_2 \leq m_1 + \frac{m_2}{1+r}$$

If ρ is real interest rate, $(1+\rho = \frac{1+r}{1+\pi})$:

$$c_1 + \frac{c_2}{1+\rho} \leq m_1 + \frac{m_2}{1+r}$$

Budget Constraint and Inflation



Rewriting equation for consumption in period 2 in the previous slide gives the budget constraint consumer faces. Again, it can be written in terms of future or present values, but both express the same constraint.

$$\frac{1+\pi}{1+r} = \frac{1}{1+\rho}$$

When inflation is low, $1 + \pi$ is close to one, so ρ approximates to $r - \pi$

$$\rho = \frac{1+r}{1+\pi} - 1 = \frac{1+r}{1+\pi} - \frac{1+\pi}{1+\pi} = \frac{1+r-1-\pi}{1+\pi} = \frac{r-\pi}{1+\pi}$$

Slope and intersection points:

Slope:

$$-\left(1 + \frac{r-\pi}{1+\pi}\right) = -\frac{1+\pi-r-\pi}{1+\pi} = -\frac{1+r}{1+\pi}$$

Budget line is flatter in the case with inflation than in the case without any inflation.

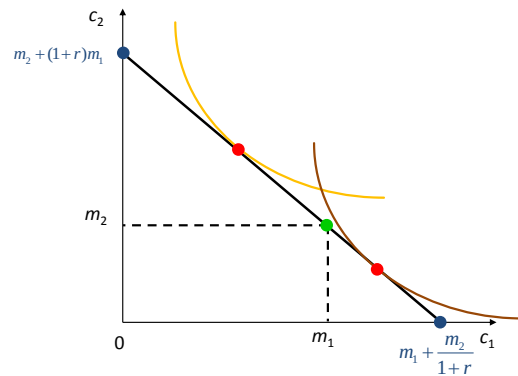
Vertical axis:

- $\frac{m_2}{1+\pi}$ is consumption you can afford from your income in period 2
- We have to incorporate the inflation as this will decrease the value of each unit of m_2 (from each dollar of m_2 you can no buy less consumption than in period 1)
- If you save m_1 in period 1, you are paid nominal interest rate r for each dollar of m_1 saved, but because price level increases (by inflation) in period 2, it is like if you earn a lower interest – real interest ρ .

Horizontal:

- If I want to borrow for consumption in period 1, I have to borrow with the current nominal interest rate. So the maximum I can borrow is $\frac{m_2}{1+r}$ and I will repay m_2 in period 2, when my consumption c_2 will be zero.
- Therefore, I don't care about inflation.

Time Preference for Consumption

Utility function: $u(c_1, c_2)$ 

Consumer with orange indifference curves: saving in optimum

Consumer with brown indifference curves: borrowing in optimum

Time Preference for Consumption and Optimum

Interior optimum:

$$\begin{aligned} & \max_{c_1, c_2} u(c_1, c_2) \\ \text{s.t. } & c_1 + \frac{c_2}{1+r} = m_1 + \frac{m_2}{1+r} \end{aligned}$$

Lagrangian:

$$\mathcal{L}(c_1, c_2, \lambda) = u(c_1, c_2) + \lambda \left(m_1 + \frac{m_2}{1+r} - c_1 - \frac{c_2}{1+r} \right)$$

FOCs:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_1} &= \frac{\partial u(c_1, c_2)}{\partial c_1} - \lambda = 0 \Rightarrow \frac{\partial u(c_1, c_2)}{\partial c_1} = \lambda \\ \frac{\partial \mathcal{L}}{\partial c_2} &= \frac{\partial u(c_1, c_2)}{\partial c_2} - \lambda \frac{1}{(1+r)} = 0 \Rightarrow \frac{\partial u(c_1, c_2)}{\partial c_2} = \lambda \frac{1}{(1+r)} \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= m_1 + \frac{m_2}{1+r} - c_1 - \frac{c_2}{1+r} = 0 \end{aligned}$$

In interior optimum:

$$\frac{MU_{c_1}}{MU_{c_2}} = 1 + r$$

We have already introduced budget constraint (for simplicity, we assume zero inflation and the same interest rate for savings and borrowings in further analysis), so let's move on to utility function. We assume classical monotonic, strictly convex preferences represented by a utility function $u(c_1, c_2)$.

Why not perfect substitutes?

- You would not care in which period you consume, 1 dollar now brings you the same utility as 1 dollar in future. But mostly, you want to have at least some consumption in both periods.

Why not perfect complements?

- It is not realistic to require always the same proportion of current and future consumption.
- If we increased consumption in only one of these periods, this would not increase your utility. Hence, assuming consumptions in different periods to be perfect complements is a bit weird.

Slope of indifference curves, if $c_1 = c_2$:

- Steep ICs mean, that consumer prefers current consumption more than future consumption. (If $|MRS_{21}| > 1$, consumer is willing to give more dollars of consumption in period 2 to get one additional dollar of consumption in period 1.)
- Flatter ICs mean, that consumer prefers future consumption more than current consumption. (If $|MRS_{21}| < 1$, consumer is willing to give less than 1 dollar of consumption in period 2 to get one additional dollar of consumption in period 1.)

In interior optimum (nonzero consumption in both periods), the individual values present consumption in terms of future consumption at its opportunity cost.

$$MU_{c_1} = (1+r)MU_{c_2}$$

- If you consume 1 dollar now, it's opportunity cost is $(1+r)$ dollars in the future.
- The last dollar used for consumption now (from which you derive utility MU_{c_1}) would bring $(1+r)MU_{c_2}$ in the future if we saved it for the future.

Corner solutions are possible:

- If $MU_{c_1} > (1+r)MU_{c_2}$, it would pay off to reallocate money from future to current period, because we value one dollar consumed now more than one dollar saved, i.e. $c_2^* = 0$, $c_1^* > 0$.
- If $MU_{c_1} < (1+r)MU_{c_2}$, it would pay off to reallocate money from current period to future, because we value one dollar saved for future more than one dollar consumed now, i.e. $c_1^* = 0$, $c_2^* > 0$.

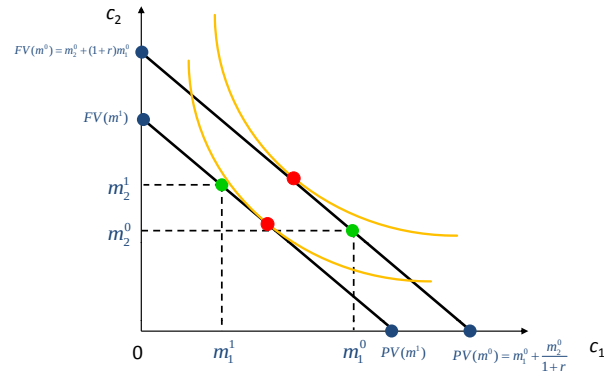
Change in Endowment

Change in endowment $(m_1^0, m_2^0) \rightarrow (m_1^1, m_2^1)$, given constant interest rate:

If $PV(m^0) > PV(m^1) \sim FV(m^0) > FV(m^1) \Rightarrow$ consumer is worse off

If $PV(m^1) > PV(m^0) \sim FV(m^1) > FV(m^0) \Rightarrow$ consumer is better off

If $PV(m^1) = PV(m^0) \sim FV(m^1) = FV(m^0) \Rightarrow$ no change in consumer's utility

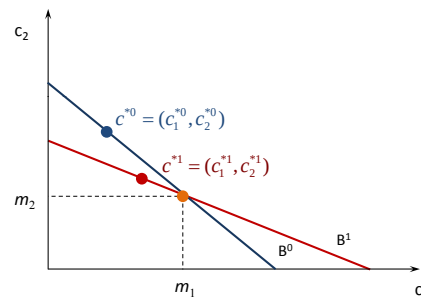


Consumption = normal good $(\uparrow PV(m) \rightarrow \uparrow c_1^*, c_2^*, \downarrow PV(m) \rightarrow \downarrow c_1^*, c_2^*)$

Decrease in Interest Rate

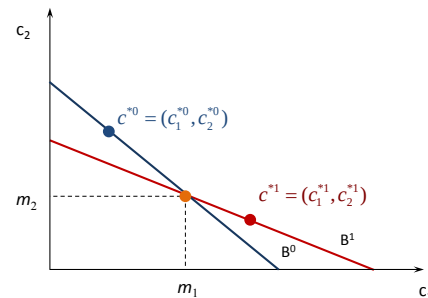
Decrease in $r \rightarrow$ flatter budget line

Lender:



If $r \downarrow$ and lender decides to remain lender, he will be worse off.

If $r \downarrow$ and lender decides to borrow money, he may be worse off or better off.



Again, we will look at how optimum changes with changes of different parameters. We start with the endowment – income received in both periods in this context.

Present and future values go line-in-line

- PV measured as an intersection point on horizontal axis, FV vertical axis

These effects are straightforward:

- If endowment changes so that a new budget line lies further from the origin, consumption in both period increases (it is a normal good) and consumer is better off.
- If endowment changes so that a new budget line lies closer to the origin, consumption in both period decreases (it is a normal good) and consumer is worse off.
- If a new endowment lies on the original budget line, amount saved or borrowed will only change without any change in consumption or utility.

The following analysis in several next slides is very similar to the analysis we discussed within the topic “Consumer buying and selling” when we looked at expected changes in utility of net buyer or seller with price change.

Slope of the budget line is $-(1+r)$, so a decrease in r leads to a flatter budget line.

Decrease in r

- it is cheaper to borrow money $\Rightarrow$ consumer can borrow higher amount in period 1 (higher c_1)
- lower interest earned on savings $\Rightarrow$ consumer can afford lower c_2

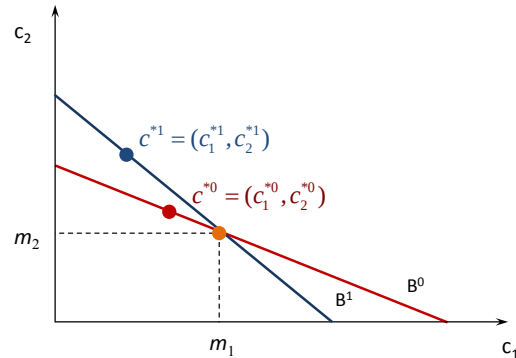
If consumer is lender, the optimum lies on the upper part of the budget line (on the left from the endowment point).

- Lender who remains lender after a decrease in r must be worse off (from revealed preferences). The new optimum was available initially and not chosen. Saved money now earns lower interest.
- If initial lender becomes borrower after a decrease in r , then using the concept of revealed preferences does not help us assess whether he is better or worse off.

Increase in Interest Rate: Lender

Increase in $r \rightarrow$ steeper budget line

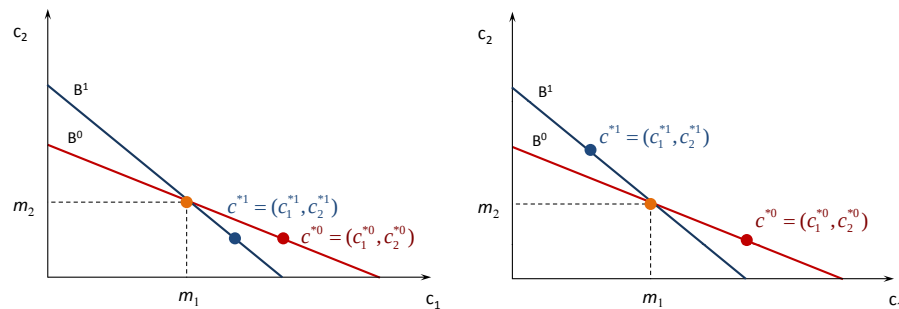
Lender:



If $r \uparrow$, lender will always remain lender, he will be better off.

Increase in Interest Rate: Borrower

Borrower:



If $r \uparrow$ and borrower decides to remain borrower, he will be worse off.

If $r \uparrow$ and borrower decides to lend money, he may be worse off or better off.

Increase in r

- more expensive to borrow money $\Rightarrow$ consumer can borrow lower amount in period 1 (lower c_1 available)
- higher interest earned on savings $\Rightarrow$ consumer can afford higher c_2

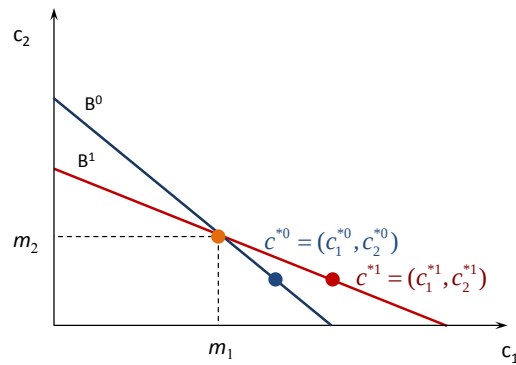
Due to the WARP, we can argue that initial lender will always remain lender after an increase in r . New budget line is steeper. Bundles lying on the right (lower) part of the new budget line were available initially and not chosen, so a bundle on the left (upper) part of the new budget line will be chosen. Consumer will be better off (initial bundle is available).

Analogical to the previous case when we discussed behavior of a lender after a decrease in r :

- If initial borrower remains borrower after an increase in interest rate, he must be worse off (using revealed preferences). New optimal bundle was available initially but not chosen.
- If initial borrower becomes lender after an increase in interest rate, then using the concept of revealed preferences does not help us assess whether he is better or worse off.

Decrease in Interest Rate: Borrower

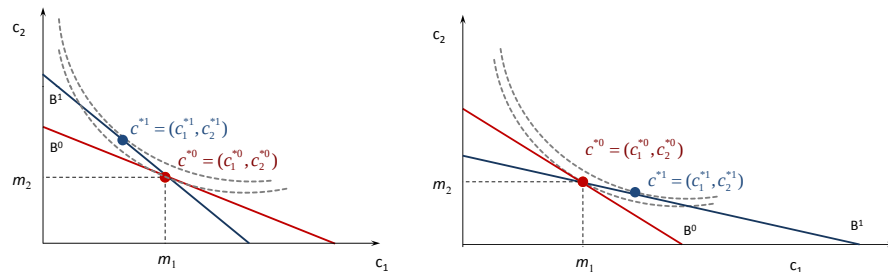
Borrower:



If $r \downarrow$, borrower will always remain borrower, he will be better off.

Change in Interest Rate

Neither borrower, nor lender:



For a given r , consumer decides to consume his endowment in each period.

- if $r \uparrow$ consumer becomes lender and he is better off.
- if $r \downarrow$ consumer becomes borrower and he is better off.

Due to the WARP, we can argue that initial borrower will always remain borrower after a decrease in r . New budget line is flatter. Bundles lying on the left (upper) part of the new budget line were available initially and not chosen, so a bundle on the right (lower) part of the new budget line will be chosen. Consumer will be better off (initial bundle is available).

If consumer decides to consume his endowment in each period, $c_1^* = m_1$ and $c_2^* = m_2$.

After a change in r , consumer will adjust his consumption so that he improves his welfare (note that his endowment is available after any change in r , so he must be better off after the change in r).

Slutsky Equation

Slutsky equation:

- how consumption in current and future periods react to changes in r

For $\pi = 0$, if BC: $c_1(1+r) + c_2 \leq m_1(1+r) + m_2$,

$$\text{Relative price of } c_1: p_1 = \frac{p_1^{FV}}{p_2^{FV}} = \frac{p_1^{PV}}{p_2^{PV}} = 1+r \Rightarrow \frac{\partial p_1}{\partial r} = 1 > 0$$

$$\text{Relative price of } c_2: p_2 = \frac{p_2^{FV}}{p_1^{FV}} = \frac{p_2^{PV}}{p_1^{PV}} = \frac{1}{1+r} \Rightarrow \frac{\partial p_2}{\partial r} = -\frac{1}{(1+r)^2} < 0$$

Slutsky Equation: Current Consumption c_1

- $\frac{\partial c_1}{\partial r} = \frac{\partial c_1}{\partial p_1} \underbrace{\frac{\partial p_1}{\partial r}}_{=1} = \frac{\partial c_1}{\partial p_1}$
- $V(m)$ is value of endowment, i.e. $V(m) = PV(m)$ or $V(m) = FV(m)$.

Slutsky equation:

$$\frac{\partial c_1}{\partial p_1} = \frac{\partial h_{c_1}}{\partial p_1} + (m_1 - c_1) \frac{\partial c_1}{\partial V(m)} \Big|_{dr=0}$$

$$\frac{\partial c_1}{\partial r} = \underbrace{\frac{\partial h_{c_1}}{\partial r}}_{<0} + (m_1 - c_1) \underbrace{\frac{\partial c_1}{\partial V(m)} \Big|_{dr=0}}_{>0}$$

$$\text{Lender: } m_1 - c_1 > 0 \quad \frac{\partial c_1}{\partial r} \leq 0 \iff \underbrace{\frac{\partial h_{c_1}}{\partial r}}_{<0} \geq \underbrace{(m_1 - c_1) \frac{\partial c_1}{\partial V(m)} \Big|_{dr=0}}_{>0}$$

$$\text{Borrower: } m_1 - c_1 < 0 \quad \frac{\partial c_1}{\partial r} < 0$$

How consumption reacts:

- We know that if r increases and consumer is lender, he remains lender, but we do not know whether he lends more money or less money.
- Or if r decreases and consumer is borrower, he remains borrower, but we do not know whether he borrows more or less money.

To answer these questions, we will derive the Slutsky equation for intertemporal choice now.

Note that we are assuming zero inflation and also same price of consumption in both periods!

Firstly, we derive relative prices of consumptions in period 1 and 2. (Remind that we used relative prices also last time when we studied how demand for leisure changes with real wage.)

- Note that $p_1^{FV} = 1+r$, $p_2^{FV} = 1$, $p_1^{PV} = 1$, $p_2^{PV} = \frac{1}{1+r}$

And then we look at how these relative prices change with interest rate r .

Remind that $PV(m) = m_1 + \frac{m_2}{1+r}$, $FV(m) = (1+r)m_1 + m_2$. It does not matter which one is used.

Substitution effect $\frac{\partial h_{c_1}}{\partial p_1} < 0$:

- $r \uparrow$: Current consumption more costly than future $\Rightarrow$ Substitute it with future c , hence $\downarrow c_1$
- $r \downarrow$: Future consumption more costly than current $\Rightarrow$ Substitute it with current c , hence $\uparrow c_1$

Income effect (notice that this effect represents both income and endowment effect altogether, but it does not have any economic sense to separate these two effects in this analysis, so we discuss them as the one income effect)

- $\frac{\partial c_1}{\partial V(m)} > 0$, consumption is a normal good, with higher income we want to consume more.
- If r increases,
 - borrower has to pay a higher interest on loan, so he is less wealthy and his consumption c_1 would decrease.
 - lender earns higher interest on savings, so he is wealthier and his consumption c_1 would increase.
- If r decreases,
 - borrower pays a lower interest on loan, so he is wealthier and his consumption c_1 would increase.
 - lender earns lower interest on savings, so he is less wealthy and his consumption c_1 would decrease.

Substitution and income effects **together**:

If r increases:	SE	IE	TE
Borrower	borrow less, $\downarrow c_1$	lower wealth, $\downarrow c_1$	$\downarrow c_1$
Lender	lend more, $\downarrow c_1$	higher wealth, $\uparrow c_1$	?
If r decreases:	SE	IE	TE
Borrower	borrow more, $\uparrow c_1$	higher wealth, $\uparrow c_1$	$\uparrow c_1$
Lender	lend less, $\uparrow c_1$	lower wealth, $\downarrow c_1$	?

Slutsky Equation: Future Consumption c_2

$$\bullet \frac{\partial c_2}{\partial r} = \frac{\partial c_2}{\partial p_2} \underbrace{\frac{\partial p_2}{\partial r}}_{<0}$$

Slutsky equation:

$$\begin{aligned} \frac{\partial c_2}{\partial p_2} &= \frac{\partial h_{c_2}}{\partial p_2} + (m_2 - c_2) \frac{\partial c_2}{\partial V(m)} \bigg|_{dr=0} \\ \frac{\partial c_2}{\partial r} &= \frac{\partial h_{c_2}}{\partial p_2} \cdot \underbrace{\frac{\partial p_2}{\partial r}}_{<0} + \underbrace{\frac{\partial p_2}{\partial r}}_{<0} \cdot (m_2 - c_2) \frac{\partial c_2}{\partial V(m)} \bigg|_{dr=0} \\ \frac{\partial c_2}{\partial r} &= \underbrace{\frac{\partial h_{c_2}}{\partial r}}_{>0} + \underbrace{\frac{\partial p_2}{\partial r}}_{<0} \cdot (m_2 - c_2) \underbrace{\frac{\partial c_2}{\partial V(m)}}_{>0} \bigg|_{dr=0} \end{aligned}$$

$$\text{Lender: } m_2 - c_2 < 0 \quad \frac{\partial c_2}{\partial r} > 0$$

$$\text{Borrower: } m_2 - c_2 > 0 \quad \frac{\partial c_2}{\partial r} \geq 0 \iff \underbrace{\frac{\partial h_{c_2}}{\partial r}}_{>0} \geq \underbrace{\frac{\partial p_2}{\partial r} (m_2 - c_2) \frac{\partial c_2}{\partial V(m)}}_{<0} \bigg|_{dr=0}$$

Slutsky Equation: Conclusions

Increase in interest rate r :*Substitution effect*

- Current consumption relatively more costly than future consumption
- Consumer should consume less in the current and more in the future period (save more, borrow less)

Income effect works differently for borrower and lender:

- Borrower
 - Has to pay higher interest in the future
 - Borrows less and consumes less in the current period
 - Future consumption may be higher (if reducing borrowing significantly) or lower (due to repayment of higher interest).
- Lender
 - Savings earn higher interest in the future
 - He is more wealthy and spends more in the future.
 - Current consumption may be lower (saving more because of earning higher interest) or higher (save less because he is more wealthy).

Substitution effect $\frac{\partial h_{c_2}}{\partial r} > 0$:

- $r \uparrow$: Future consumption less costly than current consumption $\Rightarrow$ Substitute c_1 with future c_2 , hence $\uparrow c_2$
- $r \downarrow$: Future consumption more costly than current $\Rightarrow$ Substitute c_2 with current c_1 , hence $\downarrow c_2$
- What does it mean if we say that future consumption is less costly? (less intuitive than for current consumption)
- If I want to get 12 dollars in the future I have to save 10 if interest rate is 20%, i.e. $r = 0.2$. But if interest rate increases to 50%, I have to save only 8 dollars now. So 1 dollar in the future costs me only $1/(1+r) = 1/1.2 = 0.83$ in the first case and $1/1.5 = 0.66$ in the second case.

Income effect

- If r increases,
 - borrower has to repay the loan which costs more, so he is less wealthy and his consumption c_2 would decrease.
 - lender earns a higher interest on savings, so he is wealthier and his consumption c_2 would increase.
- If r decreases,
 - borrower's costs on loan are lower, so he gets wealthier and his consumption c_2 would increase.
 - lender earns a lower interest on savings, so he is less wealthy and his consumption c_2 would decrease.

Substitution and income effects **together**:

If r increases:	SE	IE	TE
Borrower	borrow less, $\uparrow c_2$	lower wealth, $\downarrow c_2$	?
Lender	lend more, $\uparrow c_2$	higher wealth, $\uparrow c_2$	$\uparrow c_2$
If r decreases:	SE	IE	TE
Borrower	borrow more, $\downarrow c_2$	higher wealth, $\uparrow c_2$	?
Lender	lend less, $\downarrow c_2$	lower wealth, $\downarrow c_2$	$\downarrow c_2$